

A10 Claim-Bounded Black-Hole Interior Re-Exploration Archive

Synthetic Diagnostic Candidate Mapping and External Comparator Handoff
under the A10 Core Methodology

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Abstract

This manuscript reports a claim-bounded archive of a black-hole interior re-exploration route conducted under the A10 Core methodology. The work does not claim a black-hole interior solution, quantum extremal surface detection, singularity resolution, evidence for physics beyond general relativity, real stress-energy validation, numerical-relativity validation, or accepted physical candidates. Instead, the contribution is methodological and archival: previously over-pruned exploratory branches were reintroduced only as sandbox diagnostics, tested against negative controls and failure-boundary audits, and consolidated into an external-comparator handoff package. The locked route produced 216 sandbox diagnostic candidate events and two stable morphology clusters, while preserving zero physical claims and zero accepted physical candidates. Two important failures were detected and preserved: nondecaying-tail false accepts in an action/stress sandbox audit and collapse of paired morphology under an inappropriate global width gate. Both were remediated and retained as failure-boundary evidence. The final archive denies physical-entry authorization because 16 required external assets remain missing, including numerical-relativity comparators, gauge metadata, model-defined QFT/QES inputs, stress-energy/source consistency comparators, and theorem/no-go proof targets. The result is therefore best understood as a reproducible, claim-bounded diagnostic archive and specialist handoff brief, not as a report of new black-hole physics.

1 Claim boundary and scope

This paper is deliberately narrow. It reports a sandbox diagnostic archive, not a discovery of black-hole interior physics. The claim boundary is central to the result rather than a disclaimer appended after the fact. General relativity and black-hole solutions have a long, rigorous history beginning with the Einstein field equations and the Schwarzschild solution, with the Kruskal extension clarifying the maximal analytic extension of the Schwarzschild geometry [1, 2, 3]. Numerical and semi-classical approaches to black-hole physics require careful treatment of gauge, boundary conditions, constraint residuals, and high-fidelity comparators [5, 6, 7, 8]. The present archive does not meet those requirements for physical-entry claims.

The allowed claim is the following: a claim-bounded sandbox re-exploration pipeline was reconstructed under A10 Core v0.2.1; it produced a reduced diagnostic candidate map and an external-comparator handoff checklist while preserving zero physical claims and zero accepted physical candidates. The disallowed claims are equally important. This archive does not establish a black-hole interior solution, a quantum extremal surface (QES) detection, singularity resolution, beyond-GR evidence, real stress-energy validation, an Einstein-equation solution, a numerical-relativity validation, a semi-classical gravity solution, or an accepted physical candidate.

2 Background and motivation

The route was motivated by a change in A10 operating discipline. Earlier A10 black-hole exploration treated dangerous public claims and dangerous exploratory hypotheses as closely coupled. This was safe, but it may have over-pruned branches that could have been retained as non-physical diagnostics. A10 Core v0.2.1 separates entrance and exit conditions: speculative hypotheses may enter a sandbox route, but only evidence-bounded claims may exit it [13]. In this project, that distinction allowed entropy-extremal, action/stress, marching-solver, bridge-model, and handoff-comparator branches to be reopened without authorizing physical claims.

The phrase “black-hole interior re-exploration” should therefore be read as a route name, not as a claim that the black-hole interior has been physically accessed. Entanglement and extremal-surface ideas are well-established in holographic settings, from Ryu–Takayanagi surfaces to quantum extremal surfaces [9, 10]. However, this archive does not compute a model-defined QES in a controlled semi-classical spacetime. It only reopens a reduced entropy-extremal diagnostic branch as a sandbox component, then freezes a list of external assets that would be required before any physical interpretation could be attempted.

3 Methodological design

The route uses four principles.

1. **A10-free baselines first.** Flat, Schwarzschild, Kruskal, and Vaidya-surrogate checks are replayed before A10-specific hypotheses are introduced.
2. **Sandbox before physical entry.** Entropy-extremal, action/stress, bridge-model, and morphology branches are treated as reduced diagnostics, not physical candidates.
3. **Failure boundaries are evidence.** Failed runs are archived and used to strengthen guards rather than erased.
4. **External comparators are mandatory.** The route is not allowed to claim physical entry without real external assets and high-fidelity validation.

This design follows the A10 pattern of bold hypothesis entry, strict process control, and bounded public claims. The key operational distinction is that a synthetic diagnostic event is not a physical candidate. It may guide future expert comparison, but it cannot by itself justify a physical statement.

4 Route summary

Table 1 summarizes the major route blocks. The full route ledger is included in the repository as `data/R16_route_stage_ledger.csv`.

5 Locked diagnostic result

The consolidated archive inherits the R16/R17 lock:

Table 1: Condensed route summary. PASS does not imply physical-entry authorization.

Block	Purpose	Locked interpretation
R0–R1B	Re-exploration seed, A10-free base-lines, grid and perturbation audits	Known-solution scaffolds established without physical claims.
R2–R2B	Entropy-extremal diagnostic revisit and negative controls	QES-like branch reopened only as reduced diagnostic.
R3–R3C	Action/stress sandbox and failure-boundary remediation	Nondecaying-tail false accepts detected and remediated.
R4–R4B	Marching-solver and randomized holdout checks	Manufactured solver stability verified in reduced setting.
R5–R7	Candidate-map generation, adversarial controls, morphology clustering, consolidation	216 diagnostic events and two stable morphology clusters locked.
R8–R12	Pre-physical bridge, reduced residual, gauge, stress-proxy, and boundary sandbox checks	Bridge components specified but physical entry denied.
R13–R15P	External-comparator readiness, adapter/schema checks, asset binder, archive pause	16 required external assets identified; all missing.
R16–R17	Archive synthesis and repository preparation	Handoff-ready archive; not physical-entry-ready.

Quantity	Locked value
Route stage count	30
Diagnostic candidate event count	216
Stable morphology cluster count	2
Required external asset count	16
Bound required external asset count	0
Missing required external asset count	16
External comparator execution count	0
High-fidelity validation completed count	0
Physical-entry authorization	False
Physical claim count	0
Accepted physical candidate count	0

The two sandbox morphology classes are `compact_single_like` and `paired_compact_like`. These names describe diagnostic shapes in the reduced archive. They do not name physical black-hole structures. Their only current function is to specify what future external comparators would need to test.

6 Failure-boundary evidence

Two failures are especially important because they prevent the archive from becoming a confirmation-only exercise.

R3B nondecaying-tail false accepts. The action/stress holdout audit initially accepted nondecaying-tail artifacts. This failure showed that the sandbox could mistake persistent tails for admissible structure. The route was therefore remediated through a nondecaying-tail guard and an integrated rerun. The failure is preserved in the ledger as `FAIL→REMEDIATED`, not hidden.

R6 paired-morphology collapse. A later synthetic-evolution coupling preflight collapsed the paired morphology class by applying an inappropriate global width gate. The issue was not a false physical discovery but a classification failure: a two-component morphology was being judged by a single-component compactness rule. The remediation introduced morphology-specific handling of pair separation and component width.

These failures support the archive’s methodological value. They show where the route broke, why it broke, and how the guard logic was tightened without relaxing the claim boundary.

7 External-comparator handoff

The final blocker is not a vague lack of confidence. It is a concrete asset gap. The route identifies five external-comparator families:

1. known-solution reference adapters;
2. numerical-relativity metric comparators;
3. semi-classical QFT/QES comparators;
4. stress-energy/source consistency comparators;
5. singularity-consistency theorem or no-go handoff routes.

The repository lists 16 required assets across these families. They include reference datasets for flat, Schwarzschild, Kruskal, and Vaidya cases; numerical-relativity metric outputs with gauge metadata; model-defined QFT/QES calculation inputs; stress-energy/source consistency definitions; and theorem/no-go target statements. Since none of these assets are currently bound, the archive locks physical-entry authorization to false.

This is consistent with numerical-relativity practice: successful black-hole computation depends not only on code execution but also on formulation, gauge, constraints, boundary data, and validation against trusted comparators [5, 6, 7, 8]. It is also consistent with the QES literature, where the object being extremized and the entropy functional must be model-defined before a QES claim can be meaningful [9, 10, 12].

8 Discussion

The route is best interpreted through a climbing analogy. The project did not climb the wall of black-hole interior physics. It reached the end of the hiking path, mapped the cliff edge, identified the required climbing equipment, and locked a safe turnaround point. In technical terms, the synthetic route produced a reproducible sandbox diagnostic map and a handoff specification, but not a physical solution.

This distinction matters because synthetic pipelines can easily produce plausible-looking patterns. Without strict negative controls, failure-boundary preservation, gauge and boundary audits, and external comparator requirements, such patterns can be mistaken for evidence. The present archive is designed to prevent that mistake. Its value is not that it makes a new physics claim, but that it keeps the exploratory route reproducible while refusing to authorize claims that its evidence cannot support.

9 Limitations

The limitations are structural.

- No high-fidelity numerical-relativity validation has been executed.
- No real or trusted external metric dataset has been bound.
- No model-defined QFT/QES calculation has been performed.
- No real stress-energy tensor or semi-classical source has been validated.
- No theorem, no-go statement, or consistency proof for singularity behavior has been supplied.

- The diagnostic candidate map is synthetic and sandbox-limited.

Consequently, the appropriate next scientific step is not another synthetic phase. The appropriate next step is either archive/handoff publication or binding real external assets through a future `BH-R15C-bound-asset-smoke-adapter` route.

10 Conclusion

The A10-BH re-exploration archive has value as a claim-bounded diagnostic and handoff artifact, not as a black-hole physics discovery. It demonstrates that previously over-pruned exploratory branches can be reopened as sandbox diagnostics while preserving strict public claim boundaries. It consolidates 216 diagnostic candidate events into two stable morphology clusters, preserves and remediates two significant failure modes, and freezes the external assets required for any future physical-entry attempt. The archive therefore justifies GitHub repository publication and, with careful framing, may justify a preprint as a methodological case study in claim-bounded exploratory governance. It does not justify claims of black-hole interior discovery, QES detection, singularity resolution, or beyond-GR evidence.

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A Repository files relevant to this manuscript

The repository contains the following key supporting files. The manuscript should be read together with these files.

- docs/CLAIM_BOUNDARY.md
- docs/TECHNICAL_STATUS_LEDGER.md
- docs/SPECIALIST_HANDOFF_BRIEF.md
- data/R16_route_stage_ledger.csv
- data/R16_required_external_asset_checklist.csv
- static/upstream/R16_summary_lock.json